

JOAQUÍN 1.0 User Guide

Edition 1

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General Equipment Information

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1.1 General Information

Joaquín 1.0 is a computerized mechanical ventilator developed in response to the COVID-19 emergency. It is designed for use in critical care areas of hospitals that have access to medical-grade compressed air and oxygen.

Joaquín 1.0 requires constant monitoring by trained healthcare professionals. It is not a ventilator designed for outpatient emergency or patient transport.

The Joaquín 1.0 fan offers the following features:

- Display that allows monitoring the patient through graphs and numerical values in real time.
- Parameter configuration that allows for operation tailored to the patient's needs.
- Alarm settings for better patient care.
- Alarm panel capable of informing doctors in case of an emergency.
- Built-in humidifier for greater patient comfort.
- Option to take temperature readings in humidifier and patient.

- Respiratory circuits for gas delivery and exchange.
- Optional connector for temperature management in respiratory circuits.
- Design that allows for easy mobility of the equipment.

1.1.1 Operating Modes

The Joaquín 1.0 ventilator offers a wide variety of operating modes to help a diverse range of patients with different needs.

The fan has the following operating modes:

- A/C Volume
- A/C Pressure
- PSV
- CPAP (Non-Invasive)
- High Flow (Non-Invasive)

1.2 Physical description

This section provides a model of the ventilator, breathing circuits, and humidifier. Figure 1-1 shows the dimensions of the Joaquín 1.0 ventilator.



Figure 1-1. Dimensions of Joaquín fan
1.0

1.2.1 About the fan

The Figures 1-2 and 1-3 present the characteristics main from the Joaquín 1.0 fan.

Figure 1-2 shows the following elements of the fan:

1. Alarm panel.
2. Support arm.
3. Display.
4. Respiratory circuit.
5. Humidifier.



Figure 1-2. Joaquín 1.0 with accessories

Figure 1-3 shows the rear of the Joaquín 1.0 fan. From this perspective we can observe:

1. Oxygen intake.
2. Air inlet.
3. Connector for temperature sensors.
4. Cable for temperature management in respiratory circuit
5. Charger and battery storage.
6. Air and oxygen pressure regulators

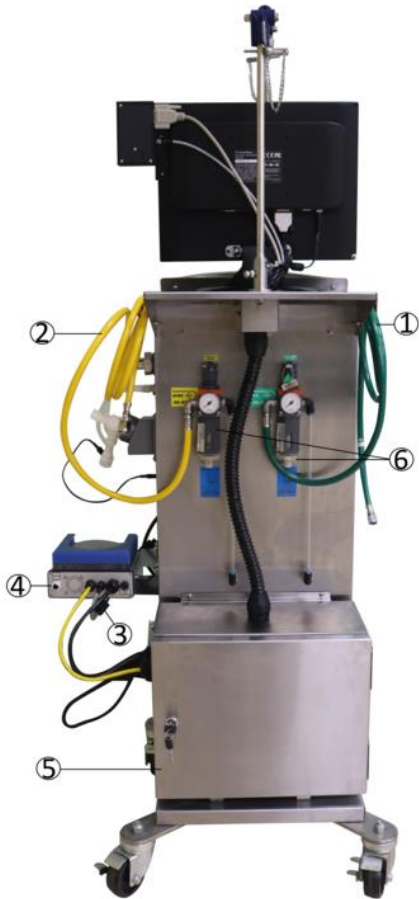


Figure 1-3. Fan back
Joaquín 1.0 with accessories

1.2.2 About the display

The Joaquín 1.0 fan has three different display models. The first model is 10 inches, the second is 15 inches, and the third is 19 inches.

The screen size used on the fan does not affect its operation in any way. Figures 1-4 through 1-6 show the displays used on the fan.



Figure 1-4. 10-inch Display



Figure 1-5. 15-inch Display



Figure 1-6. 19-inch Display

The alarm panel is located to one side of the display. This panel allows you to configure system parameters. It also generates three different alarm sounds. The alarm sound will depend on the severity of the emergency.



Figure 1-7. Monitor and alarm panel

The alarm panel features a medical-grade alarm generator that produces three distinct alarm sounds. It also includes an auxiliary alarm generator for more critical alerts.

Figure 1-8 shows the alarm panel in detail. Through this figure we can observe:

1. OLED screen for displaying messages.
2. Knob for selecting parameters in the system.
3. Medical alarm and auxiliary panel alarm.
4. Button to silence alarms.
5. LEDs to indicate system status, panel power and panel communication (see section 6.3)



Figure 1-8. Alarm Panel

Figure 1-9 shows the back of the alarm panel. The panel has an emergency power button. This button allows the PC to be restarted in an emergency when pressed and held. Its use will not stop the operation of the respirator if it is in operation.



Figure 1-9. Rear view of alarm panel

1.2.3 Accessories

The Joaquín 1.0 fan requires several accessories for proper operation. Figures 1-10 to 1-12 show the accessories used by the fan.

Figure 1-10 shows:

1. Expiration valve seal.
2. Expiration valve coupling.
3. Flow meter.
4. Hoses for flow measurement

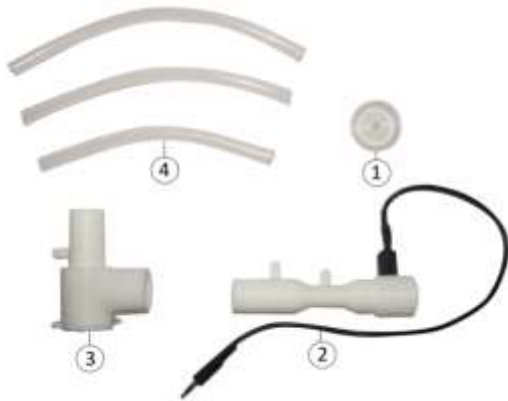


Figure 1-10. Expiration valve accessories

Figure 1-11 shows the humidifier chamber. The Joaquín 1.0 ventilator uses a Fisher & Paykel MR290 chamber. Using a different chamber does not affect the fan operation.

Joaquín 1.0 | User Manual



Figure 1-11. Humidifier chamber

Figure 1-12 shows the temperature sensing cable for the Joaquín 1.0 ventilator. This cable allows for temperature readings in the humidifier chamber, the ambient temperature, and the temperature of the gases delivered to the patient. Use of this sensor is optional. If used, it must be connected to the back of the humidifier (see section 1.4.1).



Figure 1-12. Temperature measurement cable

1.2.4 Patient respiratory circuits

Figure 1-13 shows the breathing circuit. The Joaquín 1.0 ventilator includes a Fisher&Paykel Evaqua 2 RT380 breathing circuit.



Figure 1-13. Respiratory circuit

The use of this respiratory circuit model is recommended since the ventilator incorporates connectors suitable for using the option to regulate the temperature in the circuit (see 2.4.1-2.4.2).

If you have a different breathing circuit, it can be used without any problem.

1.2.5 On team mobility

The Joaquín 1.0 fan features a robust design, constructed primarily of stainless steel. The design takes into account the unit's portability.

The fan is supported by four wheels placed at an appropriate distance from the main body to provide stability and ease of transport.

Figure 1-14 shows the fan design.



Figure 1-14. Joaquín Fan 1.0

1.3 Graphical interface of the equipment

The Joaquín 1.0 ventilator features a user-friendly interface that allows for easy operation. Figure 1-15 shows the interface in operation.



Figure 1-15. Joaquín 1.0 Fan Interface

1.3.1 General Information

The Joaquín 1.0 ventilator operates via a resistive touchscreen. The screen size can vary between 10, 15, or 19 inches. Thanks to the touch system, the user can select the desired option by pressing the device's screen. Figure 1-16 shows the main screen, which will be displayed when the device is turned on.

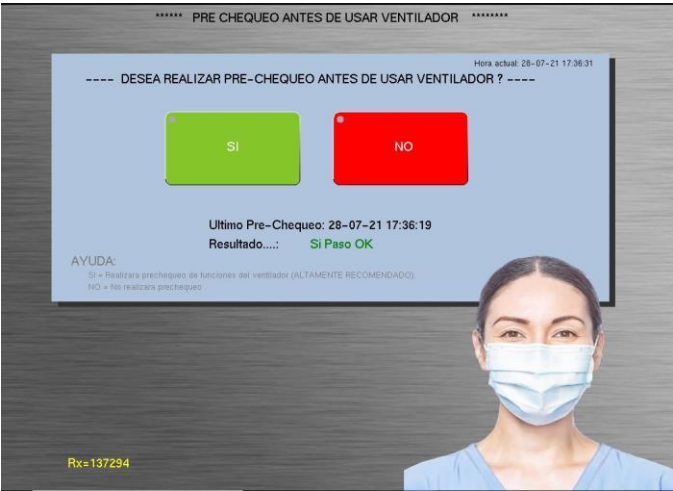


Figure 1-16. Interface home page

1.3.2 Pre-check

When you turn on the device, you will be given the option to perform a pre-check or continue without doing so. It is recommended that you perform a pre-check each time you use the fan.

Figure 1-17 shows the interface that will be displayed when the pre-check option is selected.

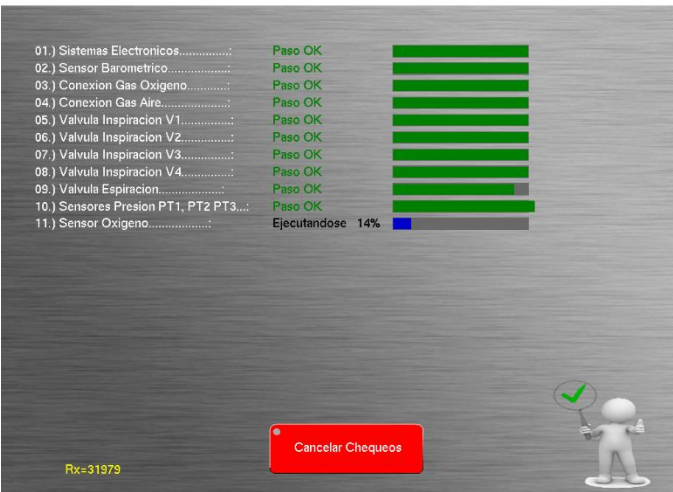


Figure 1-17. Fan pre-check.

By selecting the pre-check option, the system will evaluate all its components to ensure they are in optimal working condition. (See section 3.2)

During this process, the ventilator will give instructions to the person in charge to perform the tests correctly.

At the end of the pre-check, a message will be displayed to continue with the equipment configuration process.

1.3.3 Patient Type

After performing the pre-check or deciding not to perform it, the image shown in Figure 1-18 will be displayed. Here you will have the option to enter a new patient or continue with the previous patient.

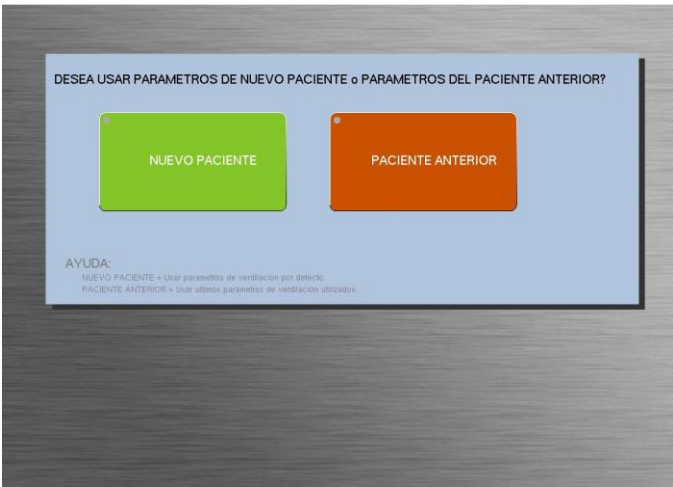


Figure 1-18. Patient selection.

If you select to enter a new patient, the system will display the options shown in Figure 1-19. In this window, you must select whether you want to enter an infant, a young person, or an adult.



Figure 1-19. Select patient type

When you select one of the options, the system will request confirmation, as shown in Figure 1-20.

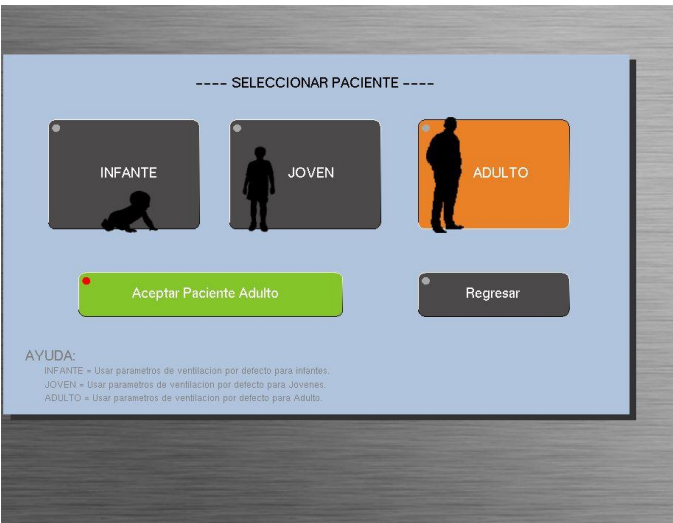


Figure 1-20. Confirm patient

Depending on the option chosen, the system will establish the maximum values that can be used to avoid harming the patient.

Once the patient type is selected, Figure 1-21 will appear on the screen. This window will also appear when selecting the “Previous Patient” option (see Figure 1-18). This window displays the following options:

- Go to ventilation screen
- Air circuit test
- Sensor Calibration

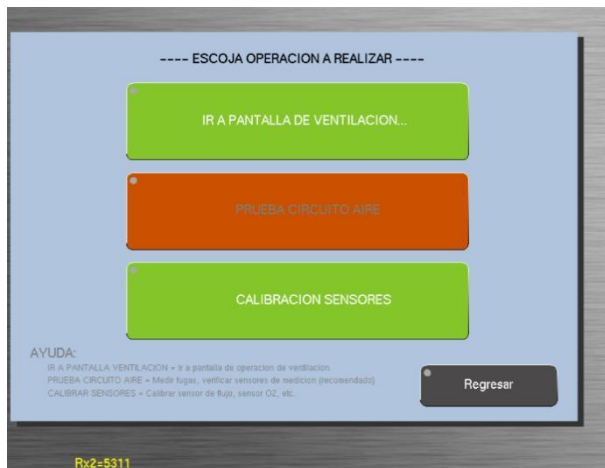


Figure 1-21. Operation start menu.

If you select the Sensor Calibration option, the following options will be displayed:

- O2 sensor
- Flow Sensor
- Expiratory Valve



Figure 1-22. Calibration Menu

Selecting one of these options will allow the system to calibrate the selected element. This can be seen in Figure 1-23.

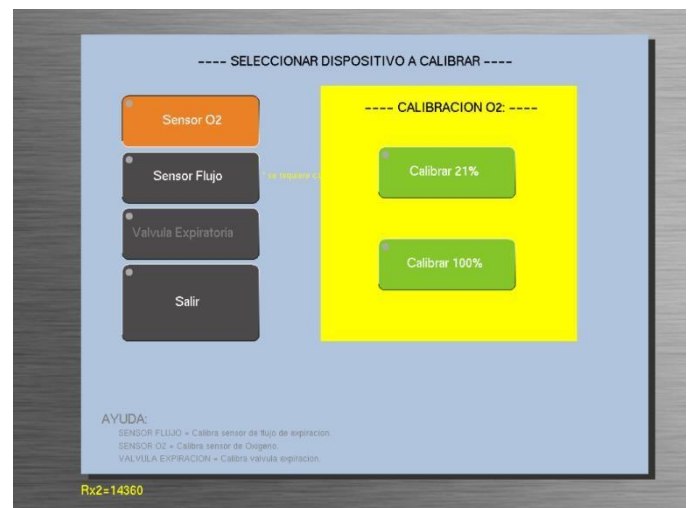


Figure 1-23. O2 Calibration

It is recommended to perform ventilator calibrations with each new patient. This procedure is detailed in section 3.4.

If the option to start ventilation is selected, Figure 1-24 will be displayed. The system will start in Standby mode and in A/C Volume operating mode.

To start operation, press the “Start” button at the bottom right (shown in 1, in Figure 1-24) and then confirm that you want the fan to start operating (Figure 1-25).

1.3.4 Parameter Selection

During the operation, it will be possible to modify the breathing parameters. To do this, select one of the parameters by tapping on it on the screen.

Next, turn the alarm panel knob. This procedure is detailed in Figure 1-26.

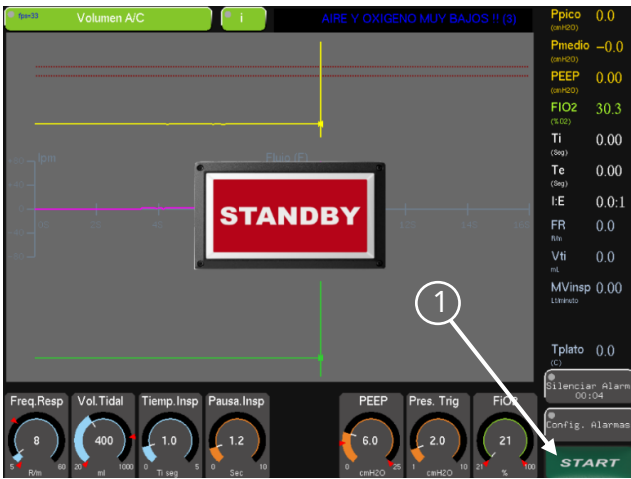


Figure 1-24. Operating Window



Figure 1-26. Parameter modification

Once the desired value has been entered, the alarm panel knob must be pressed to confirm the set value. This procedure is shown in Figure 1-27. The value can also be confirmed by pressing the parameter again on the display.



Figure 1-25. Start of operation



Figure 1-27. Parameter confirmation

1.3.5 Selection of operating mode

The Joaquín 1.0 ventilator has seven operating modes. When the ventilator starts operating, it will begin in the Volume A/C operating mode as shown in Figure 1-25.

The active operating mode can be observed at all times. This is located in the upper left corner of the screen. This is detailed in Figure 1-28.



Figure 1-28. Active operating mode

To select a different operating mode, you must pause the ventilator by placing it in Standby mode. To do this, press the Standby button in the lower right corner. This is shown in Figure 1-28. Then, confirm that you want to pause the ventilator, as shown in Figure 1-29.



Figure 1-29. Standby Mode Selection

Once the fan is in Standby mode (Figure 1-24), press the active operating mode button (shown as 1 in Figure 1-28). This will open a window where you can select a new operating mode. This is shown in Figure 1-30.



Figure 1-30. Operating mode selection

After selecting the desired operating mode, ventilation must be initiated by pressing the Start button. This procedure is shown in Figures 1-24 and 1-25.

1.3.6 Set alarms

The Joaquín 1.0 ventilator has an alarm configuration option for better patient monitoring. On the main operating screen, the "Alarm Settings" option is located in the lower right corner. This is detailed in Figure 1-31.



Figure 1-31. Select alarms

Clicking on this option will display a list of alarms which can be adjusted to the desired values. This list can be seen in Figure 1-32.



Figure 1-32. Modify alarms

Depending on the operating mode, the fan will allow you to set a greater or lesser number of alarms.

1.3.7 Monitor alarms

Once the alarms have been set, the system will begin monitoring whether any of them are active. If an alarm is active, the alarm panel will emit a continuous sound.

The sound generated by the alarm panel will depend on the alarm level (low, medium, high). Similarly, the alarm panel will display different colors on the "status" LED to indicate the alarm status.

If an alarm is active, you can view information about it by pressing the upper right corner of the screen. This procedure is shown in Figure 1-33.



Figure 1-33. Active alarms button

Clicking the active alarms button will display all active alarms with more detailed information. This window will show the date, time, duration, priority, and description of each active alarm.



Figure 1-34. Active alarms of the system

The list presents the active alarms, placing the most critical ones at the beginning and ending with those of the lowest priority.

The fan has a history log of all alarms and changes made to the fan. This log can be accessed by pressing the "i" button located at the top of the display. This procedure is shown in Figure 1-35.



Figure 1-35. Alarm and change history.

1.4 Humidifier

The fan Joaquín 1.0 has an Integrated humidifier for greater patient comfort. This maintains a temperature in the chamber to provide the patient with a gas mixture at the desired temperature and humidity.

The following sections present the features of this device.

1.4.1 General Information

Figure 1-36 shows the front of the humidifier; it is possible to observe:

1. Power button.
2. Knob for selecting parameters.
3. Resistive touchscreen for viewing and modifying parameters.
4. Plate for regulating chamber temperature.



Figure 1-36. Fan Humidifier
Joaquín 1.0

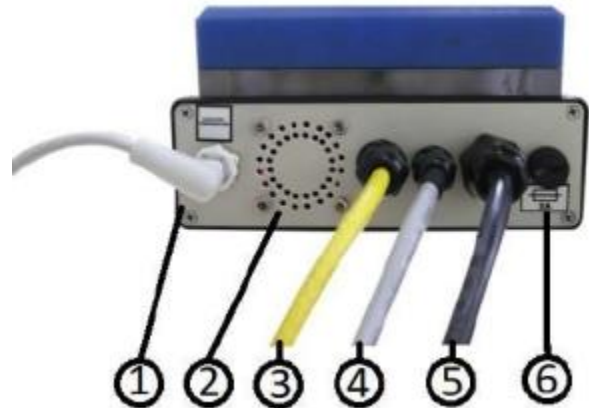


Figure 1-37. Back of humidifier

1.4.2 Accessories

The humidifier's capabilities include the ability to measure the temperature of the chamber, the ambient temperature, and the patient's temperature. A dedicated cable, shown in Figure 1-12, has been developed for this purpose.

To use this system option, simply connect the temperature measurement cable to the back of the humidifier as shown.

Figure 1-37 shows the back of the observed in Figure 1-38.

With a humidifier, it is possible to observe:

1. Temperature sensor for the camera, environment, and patient.
2. Humidifier fan.
3. Humidifier data cable.
4. Connector for respiratory circuit heaters.
5. Humidifier power cable.
6. 2A fuse.



Figure 1-38. Temperature sensor connector

Similarly, the humidifier includes a cable with connectors that can be used for the breathing circuit heaters. This is shown in Figure 1-39.



Figure 1-39. Connector for heater of ventilation circuit

To use the breathing circuit heaters, simply connect them to the cable attached to the humidifier. This process is shown in Figures 1-40 and 1-41.

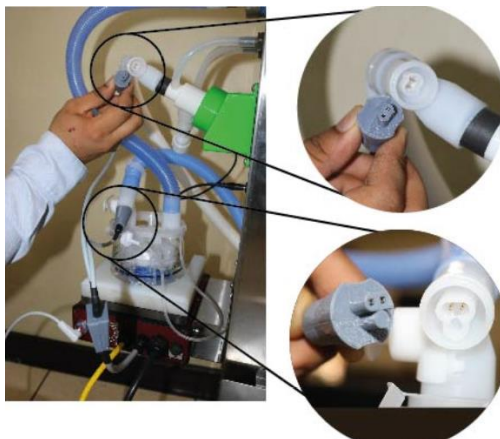


Figure 1-40. Circuit heater connection ventilation 1



Figure 1-41. Circuit heater connection ventilation 2

1.4.3 Operating Modes

The humidifier can operate automatically when its accessories are used. By correctly connecting the ventilation circuit heaters (Figures 1-40 and 1-41) and the temperature sensor cable (Figure 1-38), the humidifier will be able to regulate the temperature independently.

If the accessories are not used, the humidifier can operate using three fixed temperatures.

1.4.4 About the display

The humidifier includes a resistive touch display which allows you to view, select and modify system parameters.

When the humidifier is turned on, the screen shown in Figure 1-42 will appear. This window will remain for a few seconds while the system initializes. The operating screen will then be displayed.



Figure 1-42. Humidifier home screen

The operating window displays information about the supply voltage, plate temperature, chamber temperature, ambient temperature, and patient temperature. This can be seen in Figure 1-43.

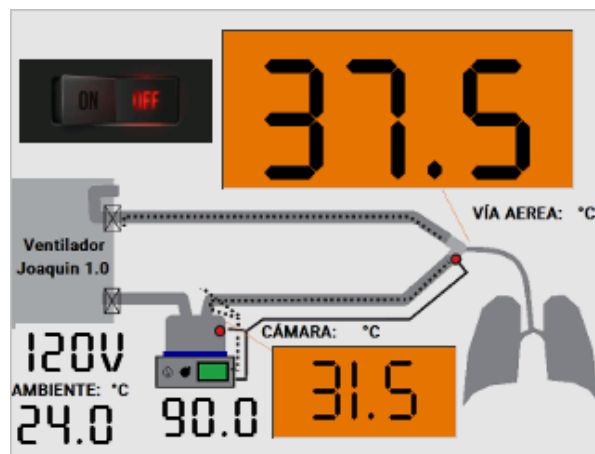


Figure 1-43. Main screen of humidifier

1.4.5 Parameter selection for automatic mode

If the temperature sensor and air circuit heating connectors are used, the humidifier can be operated automatically. When using the accessories, the desired operating parameters can be selected.

To select the parameters, press on the value you wish to modify. The humidifier allows you to modify the chamber temperature (orange color at the bottom of Figure 1-38) and the patient temperature (orange color at the top of Figure 1-38).

Pressing on the patient's temperature value will display Figure 1-44. Here, you can select the desired temperature for the patient. Once set, the humidifier will use the respiratory circuit heaters to deliver this temperature to the gases supplied to the patient.



Figure 1-44. Set temperature of the patient.

Figure 1-45 shows the window where the chamber temperature can be set. This window is displayed after pressing the chamber temperature button in the main window (Figure 1-43).

When you are on this screen, you can modify the value using the "Adjust" button shown on the screen or using the knob located on the side of the screen (Figure 1-36).

Once the desired value has been entered, press the OK button to save it. When a value is set, the humidifier will adjust the tray temperature accordingly.



Figure 1-45. Set temperature at deposit.

1.4.6 Parameter selection for manual mode

If the accessories (temperature sensor and heating cables) are not used, the humidifier can be used manually by selecting one of three preset temperatures.

To do this, you must first turn on the humidifier. It will automatically detect that the accessories are not connected. Then, you will need to press in the center of the screen.

At this point, the window shown in Figure 1-46 will appear. In this window, you will be able to select whether you want to use a low, medium, or high temperature.



Figure 1-46. Humidifier Temperatures in manual mode.

If the temperature sensor is not detected as connected, asterisks will be displayed instead of numerical values. This behavior can be observed in Figure 1-47.

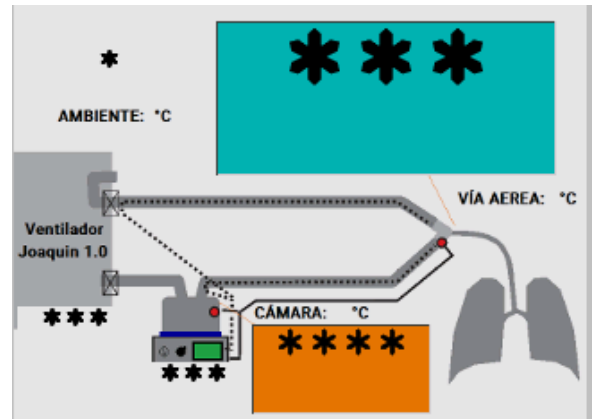


Figure 1-47. Humidifier alarms

1.4.7 Alarms

The humidifier has a built-in safety system that will trigger alarms if necessary. Because they are interconnected, the humidifier constantly sends status information to the fan. Therefore, the fan will display humidifier alarms if needed.

Alarms such as: High temperature, lack of water or some malfunction will activate the fan alarm panel and will be displayed on the monitor.

In the event of a system communication failure, the humidifier will display the image shown in Figure 1-48. If this error occurs, the system will attempt to recover automatically. If the error persists, the humidifier must be restarted.



Figure 1-48. Communication failure

Team preparation

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2.1 General Information

To use the Joaquín 1.0 fan, the following steps are required:

Task	Reference
Connect to a power source	2.2
Connect a supply of air and oxygen	2.3
Connect the circuit patient's respiratory system	2.4
Turn on the fan	2.5
Perform a pre-check of the system	3.2
Select the patient type	3.3
Perform system calibration	3.4
Select mode operation	3.5
Modify operating parameters	3.5
Modify alarms of the system	3.6

2.2 Fan power supply

Connect the Joaquín 1.0 fan to a primary power outlet. The fan is designed to accept voltages from 120V to 240V. However, a 120V power supply is recommended.

The humidifier cannot be used if a 240V power supply is used. It is recommended to check the reliability of the power outlet before connecting the device.

Figure 2-1 shows the location of the power outlet on the unit. Once connected to the power outlet, the humidifier will display the outlet voltage (see 1.4.4).



Figure 2-1. Fan power outlet.

After connecting the system, make sure to place the power cable in a secure position to prevent accidental disconnection.

2.2.1 Battery power supply

The equipment is protected by a battery bank which protects the system in case of a power outage or failure of the primary source.

The device's batteries charge whenever the fan is connected to a power source, whether or not the device is in operation.

A green LED indicator is located on one side of the fan. This light will illuminate when the batteries are fully charged; otherwise, it will indicate that the batteries are discharged or charging. The location of this indicator is shown in Figure 2-2.



Figure 2-2. Battery charge indicator

In the event that the ventilator loses its main power source, the equipment will begin operating using the battery bank without interrupting the patient's ventilation.

The device will activate an alarm which must be acknowledged to confirm that the system is operating on batteries.

If the ventilator is operating on batteries and they have a low energy level, a loud alarm will be activated to notify the doctors.

The fan has the ability to operate for up to 4 hours using batteries, provided they are fully charged.

2.3 Gas supply

The Joaquín 1.0 ventilator requires a medical-grade air and oxygen supply to operate. The air and oxygen inlets are located at the rear of the ventilator (Figure 2-3).

The ventilator incorporates two hoses with connectors suitable for the air and oxygen supplies found in hospitals (Figure 2-4). The hoses are identified using the colors yellow for air and green for oxygen.

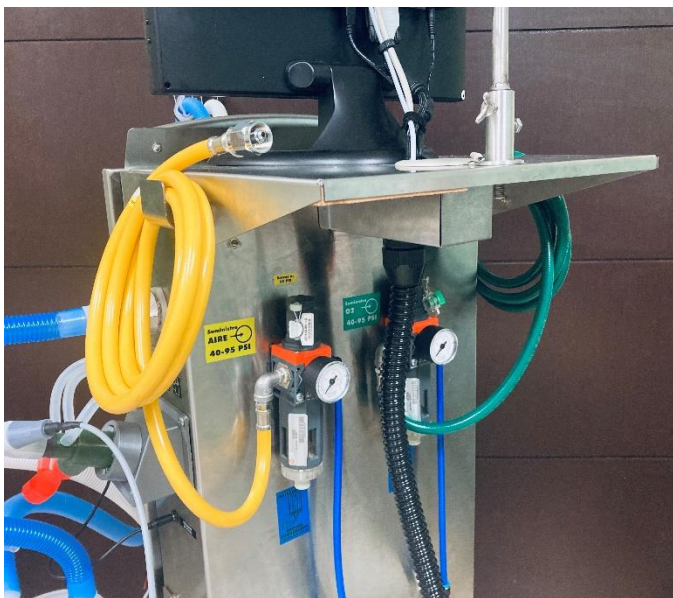


Figure 2-3. Air and oxygen inlets



Figure 2-4. Air and oxygen connectors

2.3.1 Air supply

The ventilator requires a medical-grade air supply. The inlet pressure must be in the range of 40 PSI to 94 PSI.

2.3.2 Oxygen supply

The ventilator requires a medical-grade oxygen supply. The inlet pressure should be in the range of 40 PSI to 94 PSI.

2.4 Patient's respiratory circuit

The Joaquín 1.0 ventilator includes a Fisher&Paykel Evaqua 2 RT380 breathing circuit. The use of this breathing circuit model is recommended as the ventilator incorporates connectors suitable for using the temperature regulation option in the circuit (see 2.4.1-2.4.2).

If you have a different breathing circuit, it can be used without any problem.

2.4.1 Verification of accessories

To ensure proper operation of the fan, the following accessories are required.

1. Expiration valve seal.
2. Expiration valve coupling.
3. Flow meter.
4. Hoses for flow measurement.

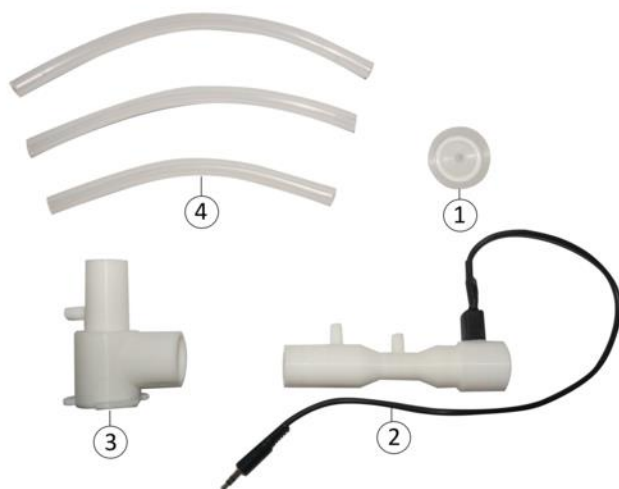


Figure 2-5. Expiration valve accessories

2.4.2 Assembly of accessories

Once the accessories are available, assembly can begin. Start by placing the exhalation valve seal, as shown in Figure 2-6.



Figure 2-6. Valve seal assembly
expiration date.

Next, the expiration valve coupling must be installed. The coupling should be positioned using the tabs on the underside of the part as a guide. Once installed, it must be rotated until it locks into a fixed position. This procedure is shown in Figure 2-7.



Figure 2-7. Expiration coupling assembly

Next, the flow meter should be placed; it should be placed on the expiration valve coupling as shown in Figure 2-8.



Figure 2-8. Flow meter assembly

Once the flow sensor is in place, it must be connected to the bottom of the expiration valve. This procedure must be performed ensuring that the connector is fully inserted. This procedure is detailed in Figure 2-9.



Figure 2-9. Flow sensor connection

Finally, the flow measurement hoses must be attached. These should be positioned as detailed in Figure 2-10.



Figure 2-10. Hose assembly for flow measurement

During the installation of the flow measurement hoses, it is essential to verify that they are properly positioned. It is necessary to check that the connectors are fully inserted into the hose, as detailed in Figure 2-11.

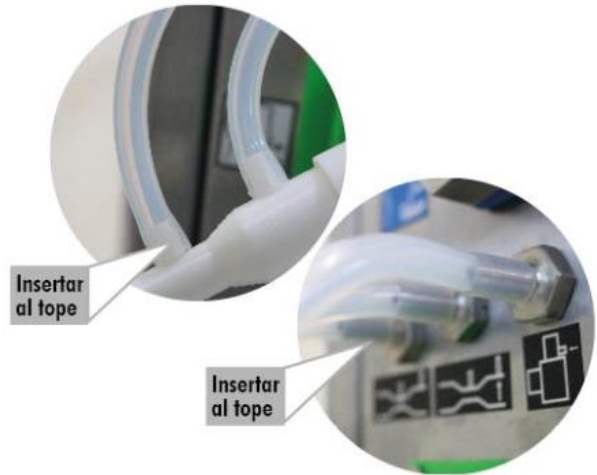


Figure 2-11. Connection verification of flow measurement hoses

2.4.3 Humidifier Chamber

To position the humidifier chamber, pressure must be applied to the plate while simultaneously moving the chamber. This procedure is detailed in Figure 2-12.

It is recommended not to put pressure on the plate with your hand as it may be at a high temperature.



Figure 2-12. Camera assembly humidifier



Figure 2-14. Camera connection humidifier with patient.

2.4.4 Connecting the breathing circuit to the ventilator

After installing the humidifier chamber, the ventilation circuit is installed. This procedure is detailed in Figures 2-13 through 2-15.



Figure 2-13. Inspiratory outlet connection with humidifier chamber.



Figure 2-15. Patient connection to valve expiration date.

2.4.5 Connection of temperature sensors

Once the breathing circuit is in place, the temperature sensor can be connected. This accessory is optional, and if it is not used, the ventilator can continue to be used normally. Figure 2-16 shows the sensor.



Figure 2-16. temperature sensor

Figure 2-17 presents the procedure to follow for connecting the temperature sensor to the humidifier.



Figure 2-17. Temperature sensor connection

Figures 2-18 and 2-19 show the procedure for placing the temperature sensor in the humidifier chamber and in the ventilation circuit.



Figure 2-18. Temperature sensor connection in humidifier chamber

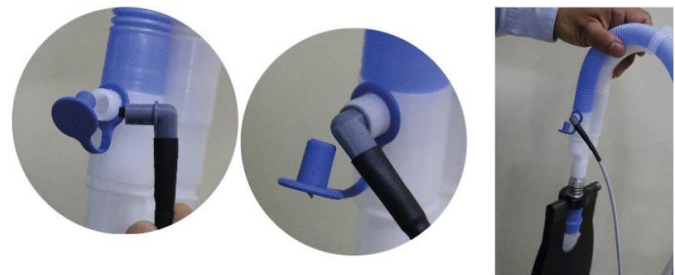


Figure 2-19. Temperature sensor connection in the ventilation circuit.

2.5 Turn on fan

Once the fan accessories are in place, you can turn on the system. To do this, activate the switch located on the lower right side of the fan.

Figure 2-20 shows the location of the switch at 1.



Figure 2-20. Fan Start-up

Setting the configuration of the fan

3.1 General Information.....	30
3.2 Pre-check.....	30
3.3 Select Patient.....	33
3.4 Sensor Calibration.....	35
3.5 Select operating mode and parameters.....	36
3.6 Setting System Alarms.....	38

3.1 General Information

Once the accessories have been attached to the fan, it can be turned on to begin operation (see Section 2.5) and the operating parameters set. This section details how to use the fan's graphical interface.

3.2 Pre-check

When the fan starts, Figure 3-1 will be displayed, offering the option to perform a system pre-check. It is recommended to perform this pre-check every time the fan is used.

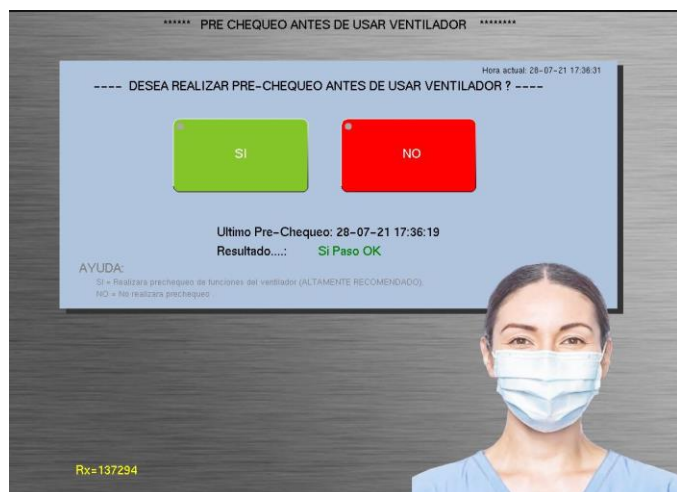


Figure 3-1. Fan initial display

Upon accepting the pre-check, the system will evaluate various elements of the fan to ensure proper functioning.

This section presents information about the checks that the system performs to ensure its operation

3.2.1 Circuit Verification

Upon startup, the system will perform a circuit verification. During this phase, the fan will examine the voltages, currents, and communication of the circuits to ensure they are functioning correctly.

If any of the circuits show negative results, the system will check if it is possible to continue operating. If the check reveals a serious fault, the fan will not allow its use.

3.2.2 Barometric sensor verification

In its second test, the ventilator will verify that its barometric sensors are functioning correctly. To do this, it will take a reading and compare the results with pre-established values. If the values obtained are within acceptable limits, the test will be considered successful.

If erroneous values are presented, the fan will check the possibility of continuing to operate.

3.2.3 Oxygen supply verification

After verifying the operation of its barometric sensors, the window shown in Figure 3-2 will be displayed.

This will require that the ventilator be connected to a source of oxygen and air. For this, it is necessary to use the ports shown in Figure 2-4 and connect them to the medical-grade oxygen and air source.

Once this procedure has been completed, it is possible to select the continue option.



Figure 3-2. Message requesting to connect outlets of air and oxygen

The ventilator will then verify that it has an oxygen supply by taking various pressure measurements. If the system is connected to an oxygen source with the correct pressure, the check will be considered successful.

3.2.4 Air supply verification

Once the oxygen supply has been verified, the system will repeat the test to ensure it has an adequate air supply. The system will take several pressure measurements at its air intake. If the system is connected to an air intake with the correct pressure, the check will be considered successful.

3.2.5 Valve verification

One of the system's key components is its air and oxygen supply valves. The Joaquín 1.0 ventilator features four Festo proportional flow control valves.

The system operates using two valves that supply oxygen, a combination valve (capable of supplying either oxygen or air), and a valve dedicated solely to supplying air. This allows for a maximum flow rate of 60 L/min when all valves are operating simultaneously.

After verifying the air and oxygen supply, the system will display a window requesting a connection between the inspiration valve and the expiration valve. This is shown in Figure 3-3.



Figure 3-3. Inspiration valve connection and expiration

After connecting the inspiration and expiration valves, the system will perform a check of the air and oxygen supply valves. The system will check each valve individually.

The system will request each valve to supply different air/oxygen flow rates and will measure the supplied flow. Each valve will be evaluated and its optimal operating condition will be determined individually.

This is one of the most important steps in the pre-check. Once the air/oxygen supply valves have been verified, the system will continue with the pre-check. In case of a failure, the system will check the option to operate by supplying a reduced flow.

3.2.6 Expiration valve verification

The system will then check the exhalation valve. For this test, the system will supply different percentages of air/oxygen. In each case, the system will measure the current drawn by the valve to determine if it is operating correctly.

If the measurements taken show the expected values, the test will be considered successful and the pre-check will proceed.

3.2.7 Pressure sensor verification

The system will then verify the system's pressure sensors. To do this, it will supply air/oxygen at different pressures and collect data. If these measurements

If the pressure matches the supplied pressure, the test will be considered successful. Otherwise, the system will check whether it can continue operating or will shut down the system.

3.2.8 Oxygen sensor verification

The system will then verify that the oxygen sensor is functioning correctly. The oxygen sensor will allow the system to deliver the oxygen percentage desired by the physician.

To perform this proof the system will supply air with varying percentages of oxygen. The system will use the oxygen sensor to verify that these percentages are correct. If these measurements match, the test will be considered successful.

If these measurements do not match expectations, the system will allow operation to continue, requesting that the sensor be calibrated.

3.2.9 Safety valve verification

The following test will verify that the safety valve is operating correctly. To do this, the system will activate and deactivate the valve while simultaneously collecting data.

The system will analyze the results to determine if the valve is functioning properly and is in a condition to operate.

3.2.10 Backup battery verification

Finally, the system will perform a test to verify that the backup batteries are functioning properly. To do this, it will request that the fan be disconnected, as shown in Figure 3-4.

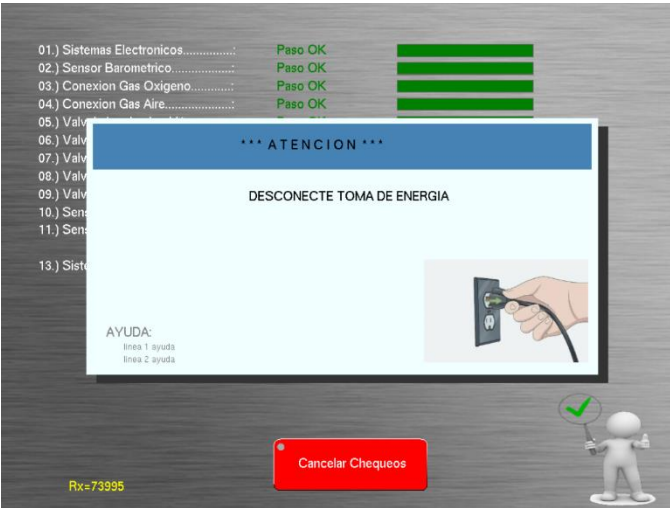


Figure 3-4. Backup battery verification

When the equipment is disconnected, the system will measure the voltages supplied by the battery and determine if it is in operating condition. Upon completion, it will request that the system be reconnected. This can be seen in Figure 3-5.

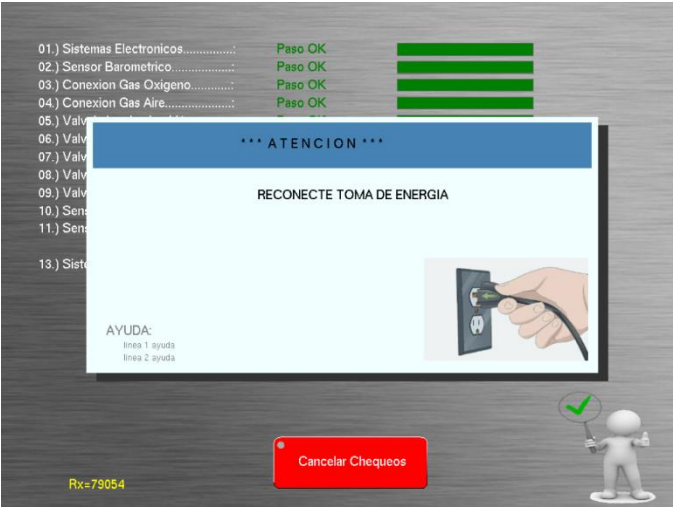


Figure 3-5. System Reconnection.

3.3 Select patient

Once the pre-check is complete, you can proceed with the necessary settings to begin ventilating a patient. The first step is to select whether the ventilator will operate on a new patient or continue operating on the previous patient. This step is shown in Figure 3-6.

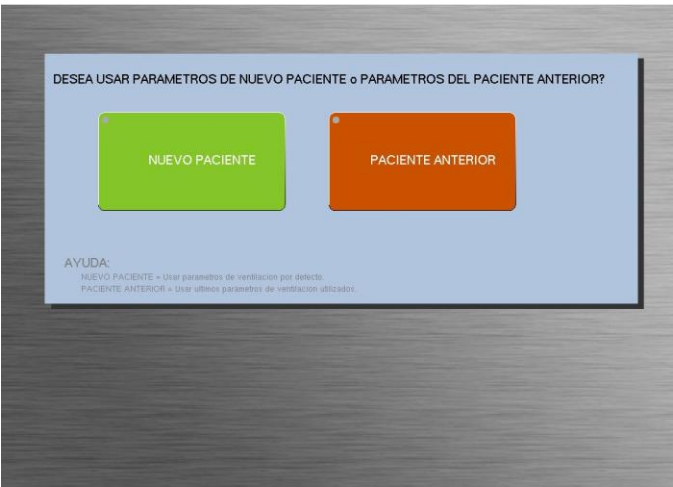


Figure 3-6. Select patient type

3.3.1 Previous Patient

If you select the "Previous Patient" option, the system will display the following options:

- Go to ventilation screen
- Air circuit test
- Sensor Calibration

When in this window, it is recommended to calibrate the sensors each time the ventilator is used with a new patient. This procedure can be found in section 3.4.



Figure 3-7. Options presented when selecting "Previous patient"

After performing sensor calibration (or if not performing calibration), you can select the "Start Ventilation" option. Selecting this option will start the fan in Standby mode (Figure 3-8). In this case, the fan will be using the last mode and parameters set.

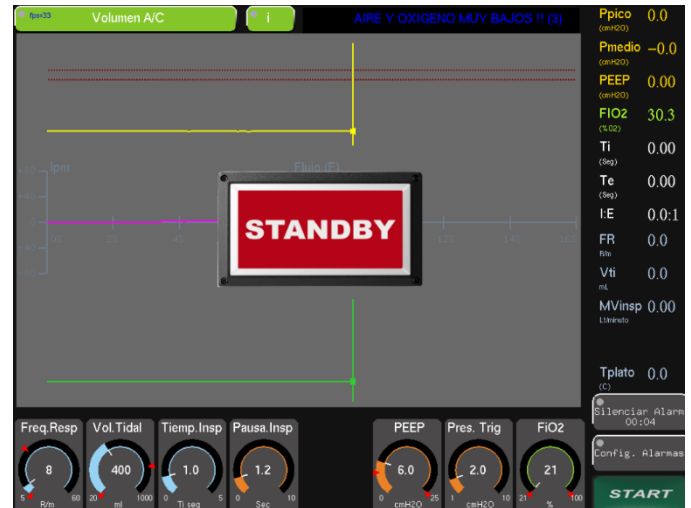


Figure 3-8. Fan in Standby

To start operation, press the "Start" button at the bottom right (shown in 1, in Figure 3-9) and then confirm that you want the fan to start operating (Figure 3-10).

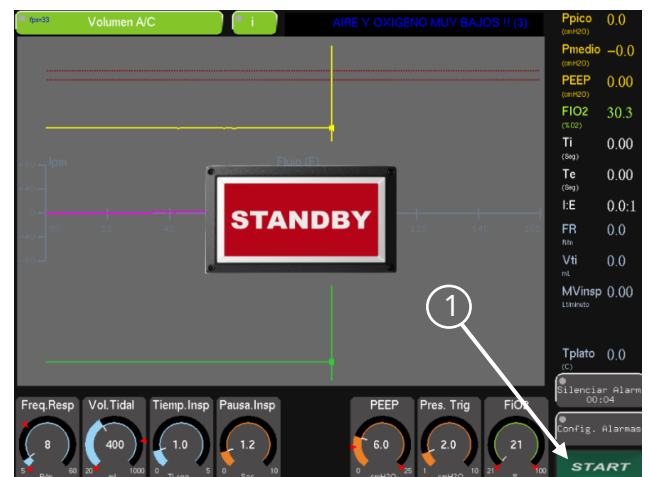


Figure 3-9. Operating Window



Figure 3-10. Start of operation

At this point the fan will begin operating in the last mode and with the last parameters used.

3.3.2 New Patient

If the "New Patient" option is selected, the system will display the option to select the patient type. These options can be seen in Figure 3-11.

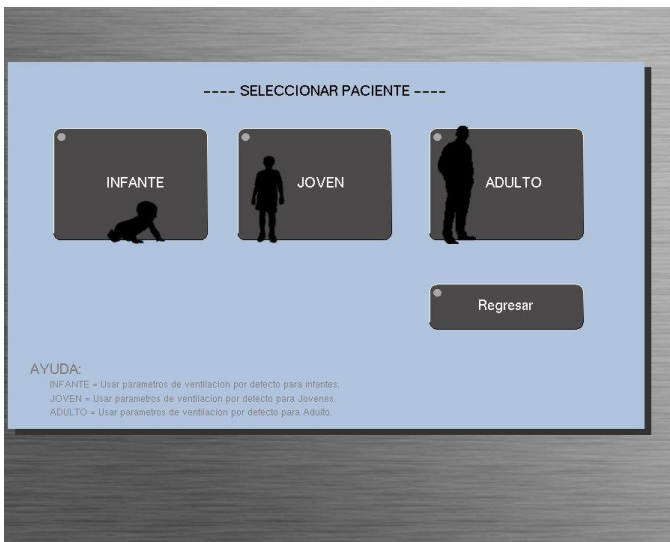


Figure 3-11. Select patient type.

When selecting the patient type, the system will ask for confirmation; after confirming, the system will display Figure 3-7.

Similarly, when in this window, it is recommended to calibrate the sensors each time the ventilator is used with a new patient. This procedure can be found in section 3.4.

When ventilation begins, the system will enter Standby mode and A/C Volume mode with predefined parameters. From this point on, it is possible to modify the operating mode and parameters. To start ventilation, follow the procedure described in section 3.3.1 or 1.3.3.

3.4 Sensor Calibration

It is recommended to select the sensor calibration option each time the ventilator is used with a new patient. Selecting this option will display Figure 3-12.



Figure 3-12. Sensor calibration

When you are in the sensor calibration window, you must select the "O2 Sensor" option to perform an oxygen sensor calibration.

Selecting this option will display two possible calibrations: Calibrate 21% or Calibrate 100%. This is shown in Figure 3-13.

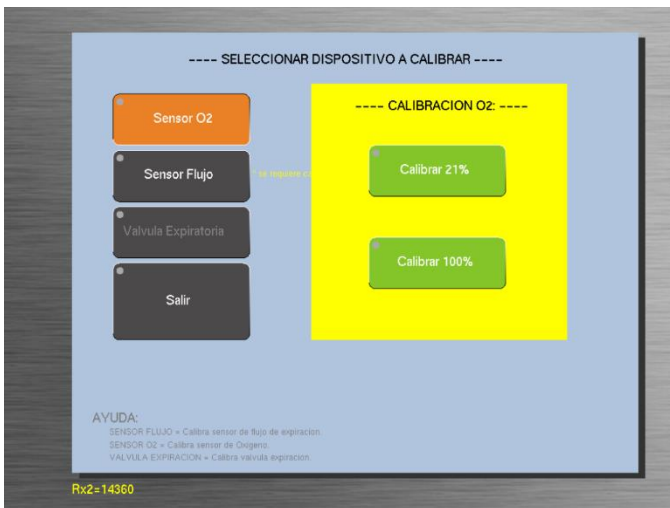


Figure 3-13. O2 sensor calibration

Selecting the Calibrate 21% option will cause the fan to use air to determine the minimum calibration value. This procedure takes approximately 1 minute. A message will then be displayed notifying you that the calibration is complete.

Next, select the Calibrate 100% option. The ventilator will then use oxygen to determine the maximum calibration value. This procedure takes approximately one minute. A message will then appear indicating that the calibration is complete.

Finally, the flow sensor must be calibrated. Selecting this option will prompt the ventilator to connect the inspiratory and expiratory valves.

Once this connection is established, you will be able to select the Start Flow Calibration option. The fan will then calibrate the sensor. This procedure takes approximately 1 minute. A message will be displayed upon completion to notify you that the calibration is finished.



Figure 3-14. Flow Sensor Calibration

3.5 Select operating mode and parameters

When ventilation starts, the fan will go into standby mode. To change the operating mode, press the upper left corner as shown in Figure 3-15.



Figure 3-15. New mode selection operation



Figure 3-16. New mode selection operation

When a new operating mode is selected, a graph and the new parameters to be used will be displayed. These parameters can be modified before confirming the new operating mode. These features are shown in Figure 3-16.

Upon confirming the new operating mode, the system will begin operating after a few seconds. Figure 3-17 shows the interface after confirming a new operating mode.



Figure 3-17. Fan operating in mode CPAP.

If the fan is running, it will not be possible to change the operating mode. The fan must be put into Standby mode to change the operating mode. To put the fan into Standby mode, follow the procedure described in section 1.3.5.

Once the desired operating mode has been selected, it will be necessary to set the appropriate parameters for the patient.

This process can be performed by pressing on the parameter you wish to modify and changing its value using the knob of the

alarm panel. This procedure can be observed in Figures 3-18 and 3-19.



Figure 3-18. Parameter Modification

Once the parameter to be modified has been selected, the desired value can be set by turning the knob and then pressing it. This procedure is shown in Figure 3-18.



Figure 3-19. Parameter Confirmation

Once the desired operating mode and parameters have been set, it is necessary to verify that the behavior is as expected using a plastic lung. Before connecting to a patient, it will be necessary to configure the system alarms.

3.6 Set system alarms

Once the fan is operating in the desired mode and with the desired parameters, it will be necessary to set the operating alarms. On the main operating page, you will find the "Alarm Settings" option in the lower right corner of the screen. This is detailed in Figure 3-20.



Figure 3-20. Select alarms

Clicking on this option will display a list of alarms that can be adjusted to the desired values. This list is shown in Figure 3-21.

Alarm values should be set in such a way that the physician can be notified in case of any undesirable behavior.

Similarly, the Joaquín 1.0 ventilator has pre-set alarms that cannot be modified. These include an alarm if the air circuit is disconnected, an alarm to notify if the device is operating on battery power, and an alarm to notify if the air or oxygen supply is not connected, among others.



Figure 3-21. Modify alarms

Depending on the operating mode, the ventilator will allow you to set a greater or lesser number of alarms. These must be configured each time the ventilator is used with a new patient.

Modos de operación

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4.1 General Information

The Joaquín 1.0 ventilator has seven different operating modes to provide the appropriate treatment for each patient. The available operating modes include:

- A/C Volume
- A/C Pressure
- PSV
- CPAP (Non-Invasive)
- High Flow (Non-Invasive)

4.2 A/C Volume

When ventilation starts, it will begin operating in Volume A/C mode by default. If the fan is operating in a different mode and you wish to return to Volume A/C mode, you must perform the following procedure.

the procedure presented in the section 3.5.



Figure 4-1. A/C Volume Mode Selection

When you select the A/C Volume operating mode, the system will display the following parameters to configure:

- Respiratory rate
- Tidal Volume
- Time for inspiration
- Inspiratory pause
- PEEP
- Trigger pressure
- FiO2

The parameters for AC Volume mode are shown in Figure 4-1. The AC Volume operating mode will provide assisted/controlled ventilation to the patient. Using the set parameters, the ventilator will deliver ventilation that meets the selected values.

He parameter Frequency Respiratory
It will set the number of breaths per minute that the ventilator will provide to the patient.

The tidal volume determines the volume of air/oxygen that the ventilator will deliver to the patient. The inspiratory time determines the duration for which the ventilator will deliver air/oxygen to the patient.

The inspiratory pause allows you to hold an inspiration for a defined time by pressing a button (section 4.2.3). At the end of a breath, the ventilator will maintain a pressure which is set by the selected PEEP value.

The ventilator will provide a mixture of air and oxygen to the patient. The FiO2 value will determine the percentage of oxygen that the ventilator will add to the gas mixture.

4.2.1 Assisted Volume Ventilation

If the patient wishes to breathe, they must apply inspiratory pressure to the system. This inspiratory pressure must exceed the value set on the device's Trigger Pressure.

The higher the trigger pressure value set, the more effort the patient will have to exert to be able to breathe on their own.

When the trigger pressure value is exceeded, the ventilator will recognize the patient's action and provide a breath using the values set in the control parameters.

When the trigger pressure is exceeded, the system will place a triangle below the breath indicator to show that the patient has established a breath. Figure 4-2 shows the triangle used to represent assisted breaths.

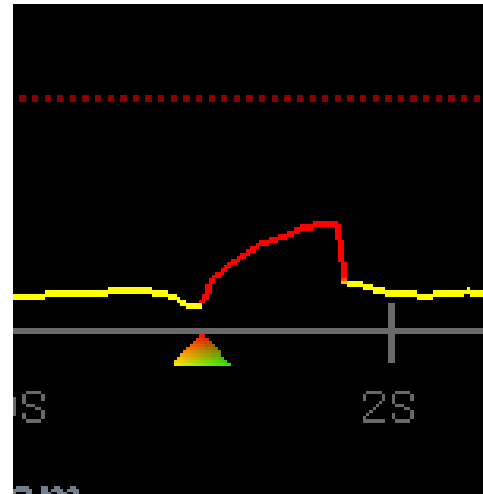


Figure 4-2. Assisted inspiration indicator

Figure 4-3 shows the interface providing volume-assisted ventilations using a calibration lung.

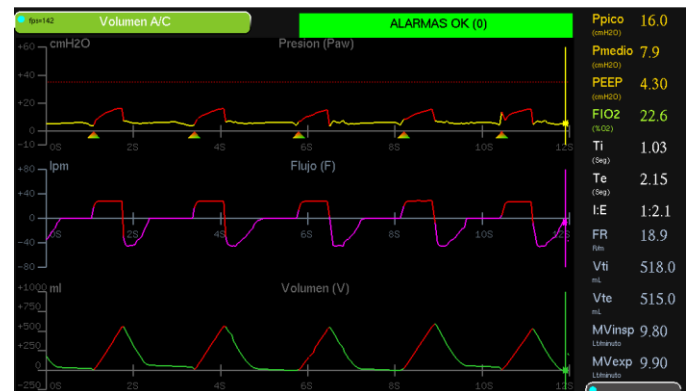


Figure 4-3. Assisted volume in operation.

4.2.2 Volume Controlled Ventilation

If the patient is unable to establish breathing patterns, the system will provide controlled ventilation. To do this, the ventilator will use the established parameters to deliver breaths to the patient at the specified time and frequency.

This operating mode will remain active until the patient requests a breath. If this occurs, it will provide the requested assisted ventilations. Otherwise, it will continue to provide controlled ventilation.

Figure 4-4 shows the system providing volume-controlled ventilation using a calibration lung.



Figure 4-4. Controlled volume in operation

4.2.3 Inspiratory Pause

When operating in A/C Volume mode, the system will offer the option to perform an inspiratory pause. This option is located at the bottom, next to the operating parameters. This can be seen in Figure 4-5.

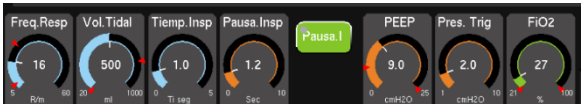


Figure 4-5. Parameters for A/C volume

To perform an inspiratory pause, you must first set the duration of the pause and then press and hold the "Pause I" button. Holding the button down will cause the system to pause the next breath. This can be seen in Figure 4-6.



Figure 4-6. Inspiratory pause.

When an inspiratory pause is performed, the average pressure value in the patient will be displayed at the bottom of the button.

4.3 A/C Pressure

When selecting A/C Pressure as the operating mode, the interface will display the following parameters to be set:

- Respiratory rate
- Time for inspiration
- Inspiratory pressure
- PEEP
- Trigger Pressure
- FiO2

As in the Volume A/C operating mode, the respiratory rate will determine the number of breaths per minute delivered to the patient.

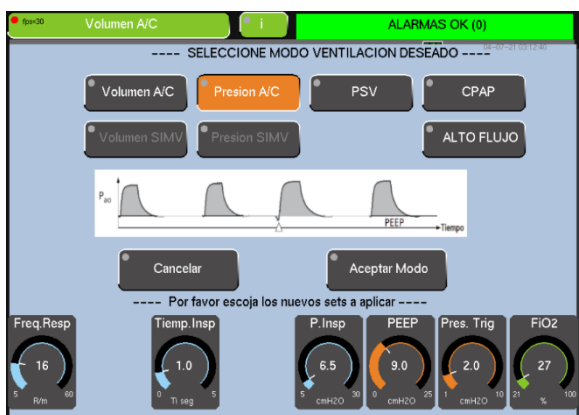


Figure 4-5. A/C Pressure Operating Mode

The inspiration time will determine the time during which air/oxygen will be supplied to the patient.

The ventilator will maintain the pressure set as PEEP at the end of each exhalation.

As with the Volume A/C operating mode, the ventilator will wait for an inspiratory pressure generated by the patient. If this inspiration exceeds the set trigger pressure, the ventilator will deliver a breath using the established parameters.

If the patient does not make any attempt at breathing, the ventilator will maintain controlled breathing using the established values.

The ventilator will supply a percentage of oxygen using the value set in the FiO2 parameter.

The A/C Pressure operating mode will attempt to achieve the set inspiratory pressure value and deliver air/oxygen to the patient, maintaining this pressure for the set inspiration time. Once this time has elapsed, the system will proceed with expiration.

Figure 4-6 shows the A/C pressure mode operating in assisted mode using a calibration lung.



Figure 4-6. Pressure assist in operation

Figure 4-7 shows the A/C pressure mode operating in a controlled manner on a calibration lung.



Figure 4-7. Controlled pressure in operation

4.4 PSV

To use PSV (Pressure by Support) mode, the system requires the following parameters to be configured:

- Raise Time
- Cycle Flow
- Time inspiration Backup
- Respiratory Rate Back Up
- Pressure Support
- PEEP
- Trigger Pressure
- FiO2

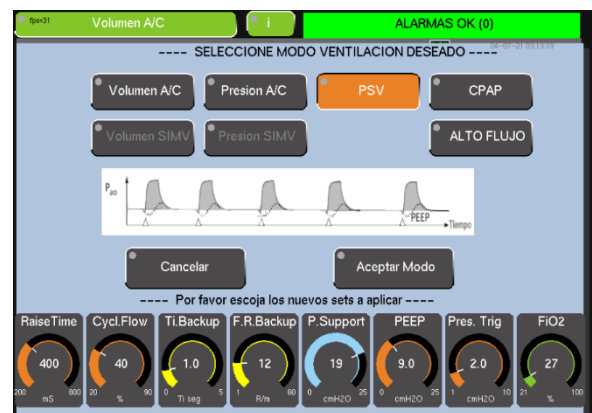


Figure 4-8 PSV Mode Parameters

The Raise time parameter will set the time it takes for the ventilator to reach the desired pressure to be delivered to the patient.

The value set as Cycle Flow will be used to determine when to exhale. Its operation will be as follows.

Once peak flow is reached, the ventilator will begin to reduce the flow delivered to the patient. Upon reaching the flow percentage set in Cycle Flow, the ventilator will begin exhalation.

Figures 4-9 and 4-10 will be used to illustrate this procedure. Figure 4-9 shows the parameters set for ventilation in PSV mode.

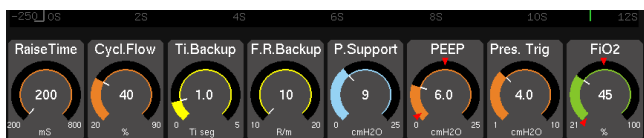


Figure 4-9. Example PSV parameters.

Figure 4-10 shows assisted ventilation provided by the Joaquin 1.0 ventilator using the parameters set in Figure 4-9.

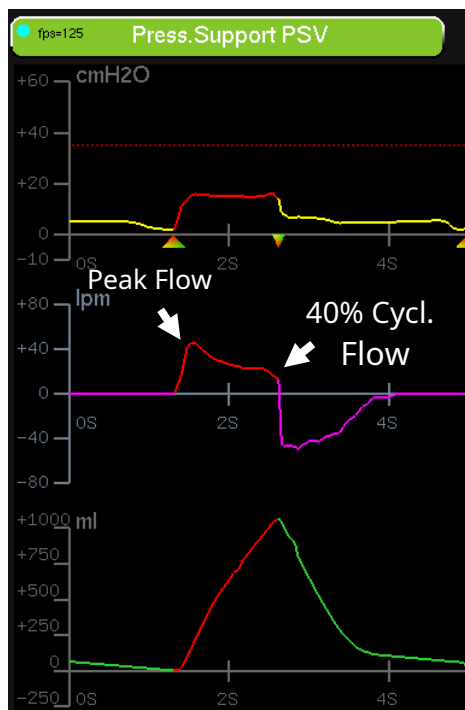


Figure 4-10. Example of PSV mode ventilation.

Upon reaching peak flow, the ventilator will begin to reduce the flow delivered to the patient. When the percentage of flow set in the Cycl. Flow parameter is reached, it will proceed with inspiration.

If the patient wishes to exhale, the ventilator will be able to sense this action, providing the patient with an exhalation even though the value set in the Cycl. flow parameter has not been met.

When the patient exhales, the ventilator will display a downward-pointing arrow as an indicator. This can be seen in Figure 4-11.

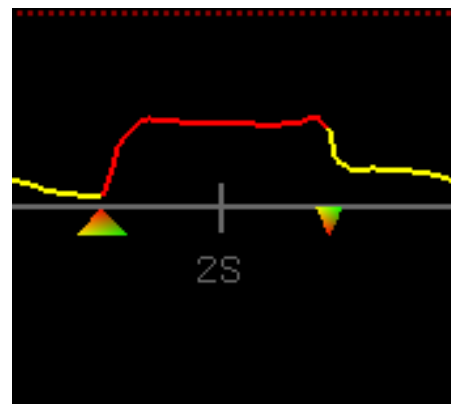


Figure 4-11. Expiration indicator

This mode of operation aims to provide the patient with assisted ventilation; therefore, the patient is expected to trigger the ventilator breath by breath.

If the patient wishes to breathe, they must apply inspiratory pressure to the system. This inspiratory pressure must exceed the value set on the device's Trigger Pressure.

When a breath is generated, the ventilator will attempt to reach the pressure value set in “P. Support”. This pressure will be maintained until the patient exhales or the Cycl. flow parameter value is reached.

At the end of exhalation, the ventilator will maintain the pressure set as PEEP in the patient.

The ventilator will supply a percentage of oxygen using the value set in the FiO2 parameter.

In both Figure 4-8 and Figure 4-9, you can see that this mode has two parameters highlighted in yellow. These parameters will be used if the mode starts operating in Backup mode.

When using this operating mode, an apnea time must be set. This can be set in the alarms section. Figure 4-12 shows the apnea setting in the Configure Alarms section.



Figure 4-12. Apnea selection

If the patient does not request any breathing for a period of time longer than that established in the apnea parameter, the ventilator will begin to operate in Pressure Back Up mode.

When entering Pressure Back Up mode, the ventilator will provide controlled breaths to the patient using the established parameters.

In this operating mode, the ventilator will use the values of the parameters Ti.BackUp to set the inspiration time and FR BackUp to set the respiratory rate.

Figure 4-13 shows the ventilator operating in PSV mode using a calibration lung.

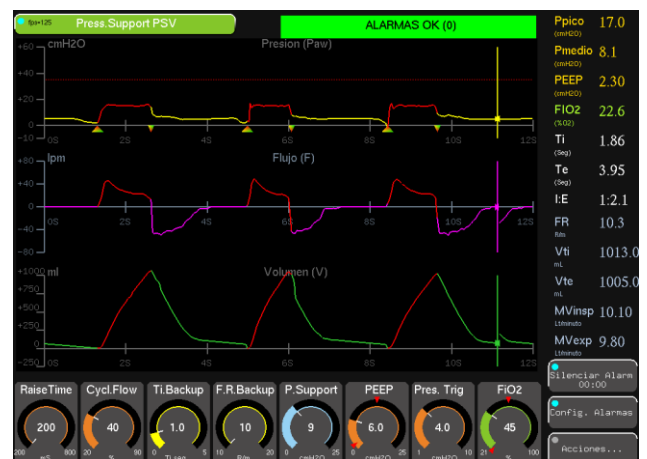


Figure 4-13. PSV mode in operation

Figure 4-14 shows the ventilator operating in PSV mode when exceeding the apnea value and working with the Backup parameters.

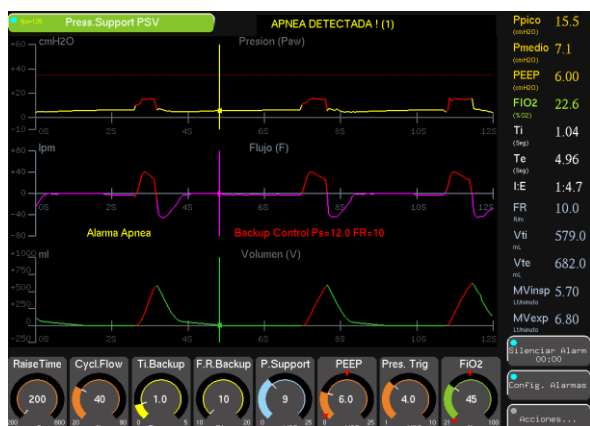


Figure 4-14. PSV mode when overcoming apnea

The Joaquín 1.0 ventilator allows for the non-invasive use of this operating mode. This mode aims to provide the patient with assisted ventilation; therefore, the patient is expected to trigger the ventilator breath by breath.

The RaiseTime and Cycl.Flow values will function in the same way as explained in PSV mode.

This operating mode will provide a constant pressure defined by the PEEP value. If the patient wishes to breathe, they must apply an inspiratory pressure to the system. This inspiratory pressure must exceed the value set on the device's Trigger Pressure.

When generating a breath, the ventilator will seek to maintain the PEEP pressure, for which it will supply a flow of air/oxygen to the patient.

The provided gas mixture will have the percentage of oxygen set in the FiO2 parameter.

The ventilator will use the same criteria explained in the PSV operating mode to perform exhalation.

As with PSV mode, an Apnea value will need to be set for this mode. This process can be seen in Figure 4-12.

4.5 CPAP

To use CPAP operating mode, the ventilator will request the following parameter settings:

- Raise Time
- Cycle Flow
- PEEP
- Trigger pressure
- FiO2



Figure 4-15. CPAP parameters

If the patient does not generate a breath after the defined apnea time, the ventilator will switch to Pressure Controlled mode using predefined values.

Figure 4-16 shows the CPAP operating mode in operation using a calibration lung.



Figure 4-16. CPAP in operation

4.6 High Flow

The High Flow operating mode will require setting the following Flow and FiO parameters².

Figure 4-17 shows the parameters to be configured in this mode.

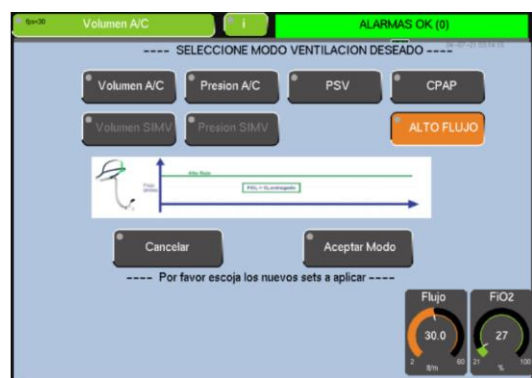


Figure 4-17. High Flow Parameters

The Flow parameter sets the flow rate that will be delivered to the patient continuously. The flow will have an oxygen concentration determined by the FiO₂ parameter. This mode will provide a constant flow of air/oxygen to the patient.

For proper operation, the humidifier will be required to provide airflow at the appropriate temperature and humidity. In addition, a nasal cannula is required, as shown in Figure 4-18.



Figure 4-18. High Flow Nasal Cannula

To use the High Flow operating mode, the ventilator's expiratory port must be connected to the humidifier. The humidifier must then be connected to the patient's cannula, as shown in Figure 4-19.

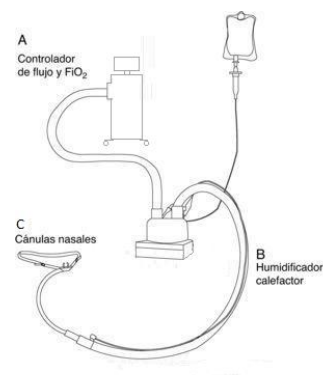


Figure 4-19. High Flow Connection

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5.1 General Information

The Joaquín 1.0 ventilator uses its display to show information about the breaths delivered to the patient. This chapter will show the different data displayed by the ventilator.

5.2 Patient numerical data

During operation, the ventilator will display numerical data that will allow the physician to monitor the equipment's performance. The data displayed includes:

- Peak Pressure
- Average pressure
- PEEP
- FiO2
- Time for inspiration
- Expiration time
- I:E
- Respiratory rate
- Tidal volume of inspiration
- Inspiratory volume per minute
- Humidifier status
- Temperature of the gases supplied to the patient

This data will be displayed on the right side of the screen and will update with each breath taken. Figure 5-1 shows an example of the displayed data using a calibration lung.

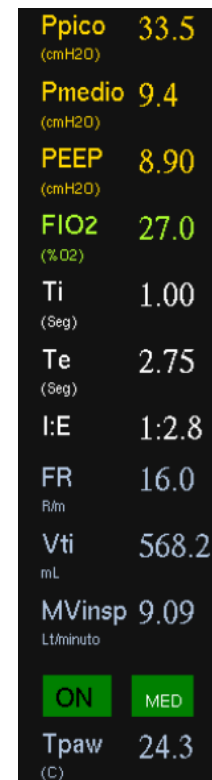


Figure 5-1. Ventilation data.

5.3 Patient graphic data

In addition to the numerical data, the ventilator will display graphs that will allow you to observe the behavior of each breath.

Pressure, flow, and volume graphs will be displayed for all operating modes.

5.3.1 Pressure

The ventilator will use pressure sensors built into its main circuit to collect data on the pressure delivered to the patient.

The collected data will be displayed in yellow at the top of the screen. During inspiration, the system will display the pressure delivered to the patient in red.

Similarly, a red line will appear at the top of the graph. This line indicates the maximum pressure value. This can be adjusted in the "Config. Alarms" section.

If the maximum pressure value is reached, the ventilator will open its safety valve to prevent harm to the patient. Figure 5-2 shows an example of the pressure graph using a calibration lung.

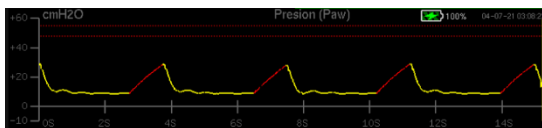


Figure 5-2. Pressure Graph

5.3.2 Flow

The fan will use information from its valves and flow sensor to collect the data required to generate the flow graph. This graph will be displayed below the pressure graph.

The flow graph will be displayed in purple. As with the pressure graph, the inspirations provided by the ventilator will be displayed in red. Figure 5-3 presents an example using a calibration lung.

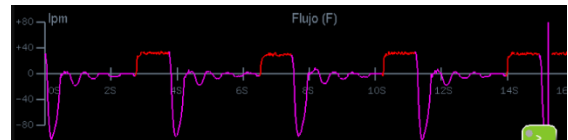


Figure 5-3. Flowchart

5.3.3 Volume

The system will use its internal sensors to determine the volume delivered to the patient. These values will be used to generate the volume graph, which will be displayed below the flow graph.

The volume graph will be displayed in green. As with the flow and pressure graphs, the inspirations delivered by the ventilator will be displayed in red. Figure 5-4 shows an example using a calibration lung.

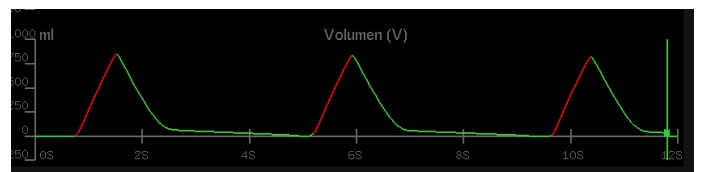


Figure 5-4. Volume Graph

5.4 Modify operating parameters

The fan will display the current operating mode parameters at the bottom of the display. These parameters can be modified by pressing the desired parameter on the display and then turning the alarm panel knob (see Section 3.2). Figure 5-5 shows an example of the current system parameters.

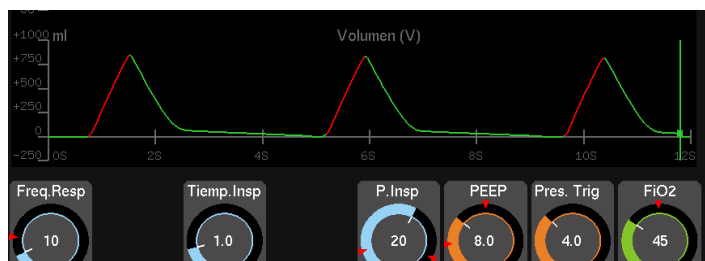


Figure 5-5. Operating parameters

When selecting a new usage mode, the parameters of the new mode will be displayed before you confirm its use. You can modify the parameters of the new mode before confirming its use. Figure 5-6 shows an example of this procedure.



Figure 5-6. Operating parameters

5.5 Modify alarm parameters

In the lower right corner of the display, you will find the "Alarm Settings" button. Pressing this button will open the alarm menu. The alarm menu is shown in Figure 5-7. In this menu, you can set the desired alarm values.



Figure 5-7. Alarm configuration

To set an alarm value, press the desired value on the display and then turn the alarm panel knob. Once the desired value is reached, press the alarm panel knob again (See Section 3.2).

5.6 Alarm history and active alarms

If one or more alarms are active, you can view information about them by pressing the upper right corner of the screen. This procedure is shown in Figure 5-8.



Figure 5-8. Active alarms button

Clicking the active alarms button will display all active alarms with more detailed information. This window will show the date, time, duration, priority, and description of each active alarm, as shown in Figure 5-9.



Figure 5-9. Active alarms of the system

The list presents the active alarms, placing the most critical ones at the beginning and ending with those of the lowest priority.

The fan has the option to view a history of all alarms and changes made to the fan. This can be viewed by pressing the "i" button located at the top of the monitor. This procedure is shown in Figure 5-10.



Figure 5-10. Alarm and change history.

5.7 Analysis of values in graphs

The fan allows data capture for analysis by pressing the freeze button located on the right side of the monitor. This is shown as 1 in Figure 5-11.

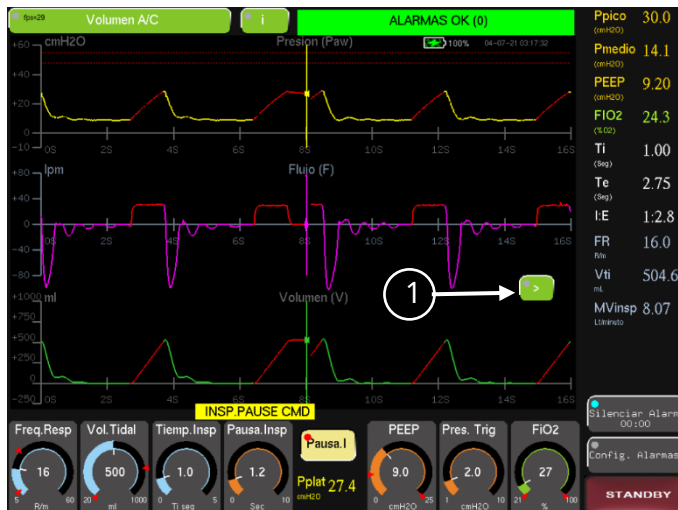


Figure 5-11. Freeze Image Button

Pressing the freeze button allows the system to pause the graphs so that the values they contain can be examined in detail. This can be seen in Figure 5-12.



Figure 5-12. Image of frozen graphs

When the graphs are frozen, four boxes will appear displaying the data values. The white box will show the pressure values, the gray box the time values, the purple box the flow values, and the green box the volume values.

The system allows you to scroll through the graphs by turning the knob on the alarm panel. This can be seen in Figure 5-13.



Figure 5-13. Data analysis using a knob alarm panel

5.8 Humidifier Monitoring

Because the humidifier and the fan are connected, they can communicate with each other. This allows information from the humidifier to be displayed on the fan's monitor.

By observing the ventilator data displayed on the right side of the monitor, it is possible to observe the humidifier status and the patient's temperature. This can be seen in Figure 5-14.



Figure 5-14. Humidifier data deployed in fan

This data will be displayed for all operating modes as long as the humidifier is turned on. Pressing on this data will display more information about the humidifier, as shown in Figure 5-15.

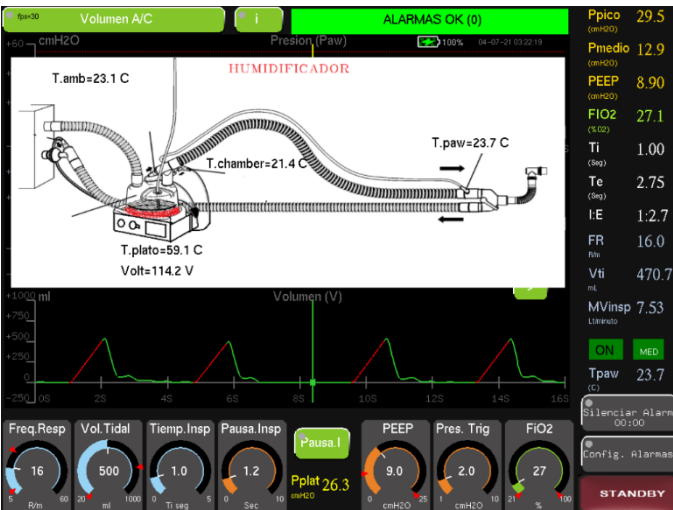


Figure 5-15. Humidifier status

Whenever the humidifier is switched on, it will be possible to observe the temperature values of the plate and the voltage in the system.

If the temperature sensor is used (Sections 1.2.3 and 1.4.2), it will be possible to observe the temperature values of the gases in the chamber, the environment, and the patient. This is shown in Figure 5-15.

If breathing circuit heaters are used, the system will display them as shown in Figure 5-16.

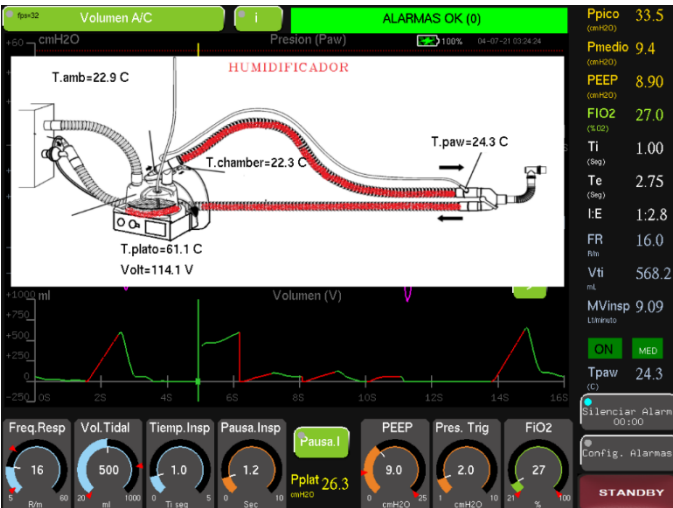


Figure 5-15. Humidifier status using breathing circuit heaters.

6 Alarms

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6.1 General Information

The Joaquín 1.0 fan uses an alarm panel to notify users of alerts related to system operation. This chapter presents information related to system alarms.

6.2 Alarm Panel

The alarm panel was designed taking into account all necessary safety factors. It is powered by the fan, as well as an internal battery bank.



Figure 6-1. Alarm Panel

When the fan is turned on, the alarm panel will establish communication with the main circuits. Both the panel and the main circuits will maintain communication at all times.

The panel features an OLED screen to display messages when needed. It also includes two speakers to generate audible alarms for medical personnel.

A medical-grade alarm generator was used which incorporates three different alarms: low, medium, or high alarm.

The alarm panel also includes three LEDs to display alarms. These are known as: Status, Power, and Communication alarms.

6.3 Alarm status, power and communication

The alarm panel has three LEDs on its front. These will be used to display visual alarms.

The Status LED indicates the fan's status. If there are no active alarms, this LED will remain lit in green.

In case of a low alarm, the Status LED will light up in blue and will continue flashing on and off at a low frequency.

In the event of a medium alarm, the Status LED will light up in yellow and will continue to flash on and off at a medium frequency.

In case of a high alarm, the Status LED will light up in red and will continue flashing on and off at a high frequency.

The alarm panel has its own battery bank, allowing it to operate independently of the fan. When the fan is turned on, the panel receives power from it, which charges the batteries.

As long as the fan is on, it will charge the alarm panel's batteries. Similarly, as long as the fan is on, it and the alarm panel will maintain communication.

If the alarm panel loses power from the fan, the Power LED will illuminate, indicating that the panel does not have a primary power source and is operating on battery power. At this time, the panel will also activate an audible alarm.



Figure 6-2. Energy loss in Panel of Alarms

To silence the alarm, you must acknowledge it by pressing the alarm silence button on the panel. Pressing the button will silence the audible alarm, but the Power LED will remain illuminated. If the primary power supply is restored, the LED will turn off, and the panel will resume normal operation.

If its primary power source is not restored, the panel will turn off.

As mentioned previously, the fan will maintain communication with the alarm panel while it is running. If the panel loses this communication at any point, an audible alarm will sound and the communication LED will illuminate.



Figure 6-3. Loss of communication in Panel of Alarms

To silence the alarm, press the Mute Alarm button on the panel. Pressing this button will silence the audible alarm for two minutes, and the communication failure LED will remain illuminated.

If the two minutes expire and no communication is received from the fan, the alarm will be activated again.

6.4 Low, medium and high alarm

The fan will be able to activate three different audible and visual alarms.

When a low alarm is activated, the alarm panel will display the Status LED flashing blue while emitting a two-tone sound. This sound will repeat every 30 seconds until the alarm is acknowledged.

If a medium-level alarm is detected, the alarm panel will display a flashing yellow Status LED while emitting a three-tone sound. This sound will repeat every 7.5 seconds until the alarm is acknowledged.

For a high alarm, the alarm panel will display the Status LED flashing red while emitting a 5-tone sound. This sound will repeat every 2.5 seconds until the alarm is acknowledged.

6.5 Alarm Volume Adjustment

The Joaquín 1.0 fan does not have the option to adjust the volume of active alarms. Low, medium, and high alarms will all activate at the same volume. These will generate tones between 85 and 95 dB.

6.6 Temporarily silence alarm

When an alarm is active, it can be silenced by pressing the mute alarm button on the alarm panel or by pressing the mute alarm button on the display. Both are shown in Figure 6-4.



Figure 6-4. Alarm mute buttons

Pressing either button will silence active alarms for two minutes. If the problem persists after two minutes, the alarm will reactivate. If the problem is resolved within those two minutes, the alarm will remain silenced.